



National Textile University

Department of Computer Science

Subject:

Mobile application and development

Submitted To:

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Registration No:

23-NTU-CS-1221

Lab no:

6

Semester:

6th

1. Create an **application** that has Icon button with toggle Icon that changes the **theme** of your application as **dark or light theme**. Create the button on the App bar.

Hint: Use ternary Operator

2. Add some paragraphs of text on the screen and **use the theme data class** to implement the **textTheme** properties like **font style, font size, and colors** on the text.

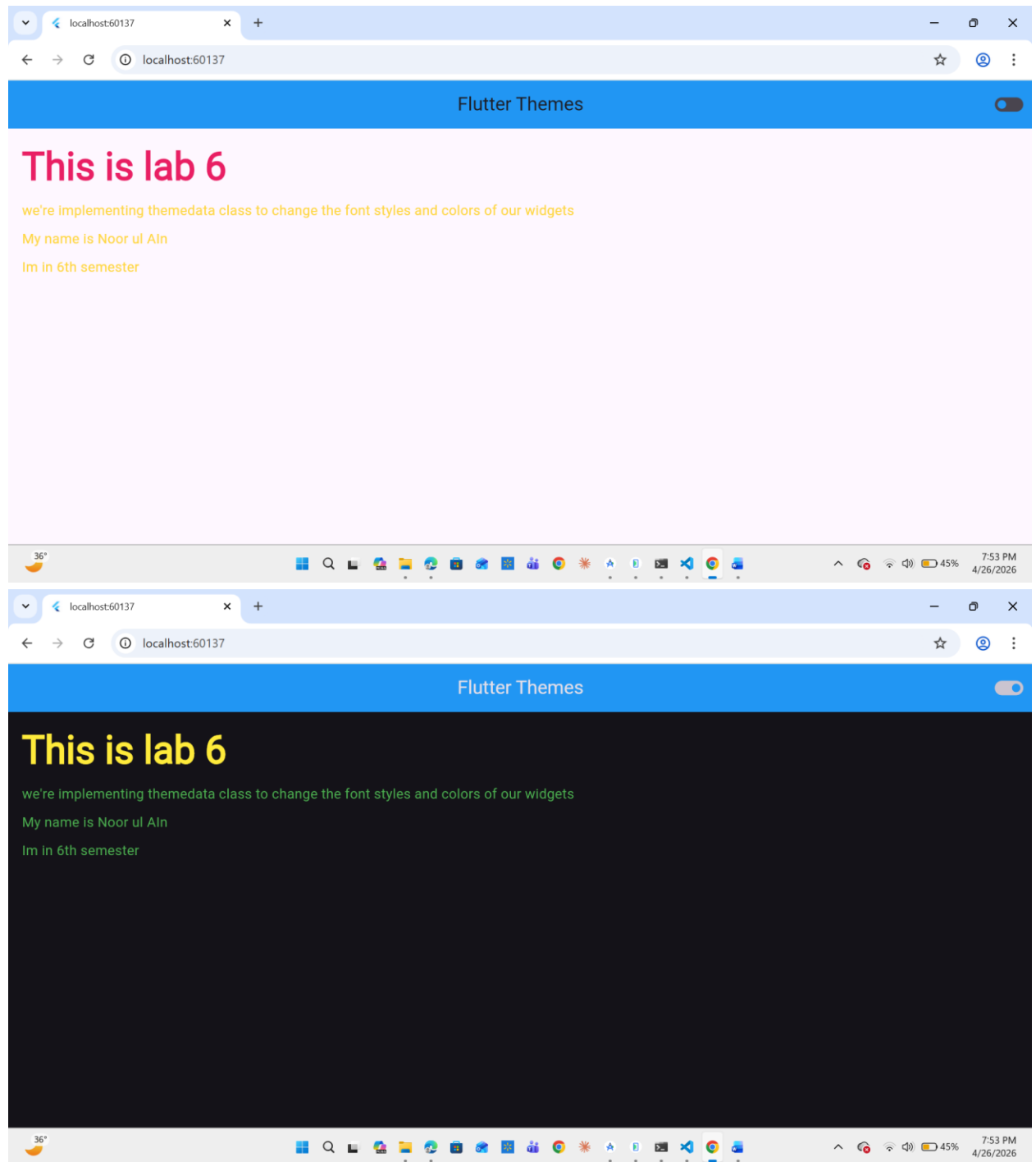
CODE:

```

1 import 'package:flutter/material.dart';
2 void main() {
3   runApp(const MyWidget() );
4 }
5
6 class MyWidget extends StatefulWidget {
7   const MyWidget({super.key});
8   @override
9   State<MyWidget> createState() => _MyWidgetState();
10 }
11 class _MyWidgetState extends State<MyWidget> {
12   bool isLight=true;
13   @override
14   Widget build(BuildContext context) {
15     return MaterialApp(
16       theme:isLight ? ThemeData.light().copyWith(
17         textTheme: const TextTheme(
18           headlineMedium: TextStyle(
19             fontSize: 45,
20             fontWeight: FontWeight.bold,
21             color: Colors.pink,
22           ),
23           bodyMedium: TextStyle(
24             fontSize: 16,
25             color: Colors.amberAccent, ) ),
26       ThemeData.dark().copyWith(
27         textTheme: TextTheme(
28           headlineMedium: TextStyle(
29             fontSize: 45,
30             fontWeight: FontWeight.bold,
31             color: Colors.yellow ),
32           bodyMedium: TextStyle(
33             fontSize: 16,
34             color: Colors.green, ) ),
35       ),
36       debugShowCheckedModeBanner: false,
37       home:Builder(
38         builder: (context) {
39           return Scaffold(
40             appBar: AppBar(
41               backgroundColor: Colors.blue,
42               title:Text("Flutter Themes").centerTitle: true,
43               actions: [
44                 IconButton(
45                   onPressed: (){
46                     setState(() {
47                       isLight=!isLight;
48                     });
49                     icon:Icon(isLight?Icons.toggle_off:Icons.toggle_on),
50                     iconSize: 40, ) ],
51             body: Padding(
52               padding:EdgeInsets.all(16.0),
53             child:Column(
54               crossAxisAlignment: CrossAxisAlignment.start,
55               children: [
56                 Text("This is lab 6 ",style: Theme.of(context).textTheme.headlineMedium),
57                 SizedBox(
58                   width: 10,
59                   height:10,
60                 ),
61                 Text("we're implementing themedata class to change the font styles and colors of our widgets ",
62                 ),
63                 SizedBox(
64                   width: 10 height:10,
65                 ),
66                 Text("My name is Noor ul AIIn ", ),
67                 SizedBox(
68                   width: 10 height:10,
69                 ),
70                 Text("Im in 6th semester",),
71               ],)
72             /* child: Text("HeLlO WORLD"), */
73           ),
74         ),
75       );
76     } )
77   );}
78 }

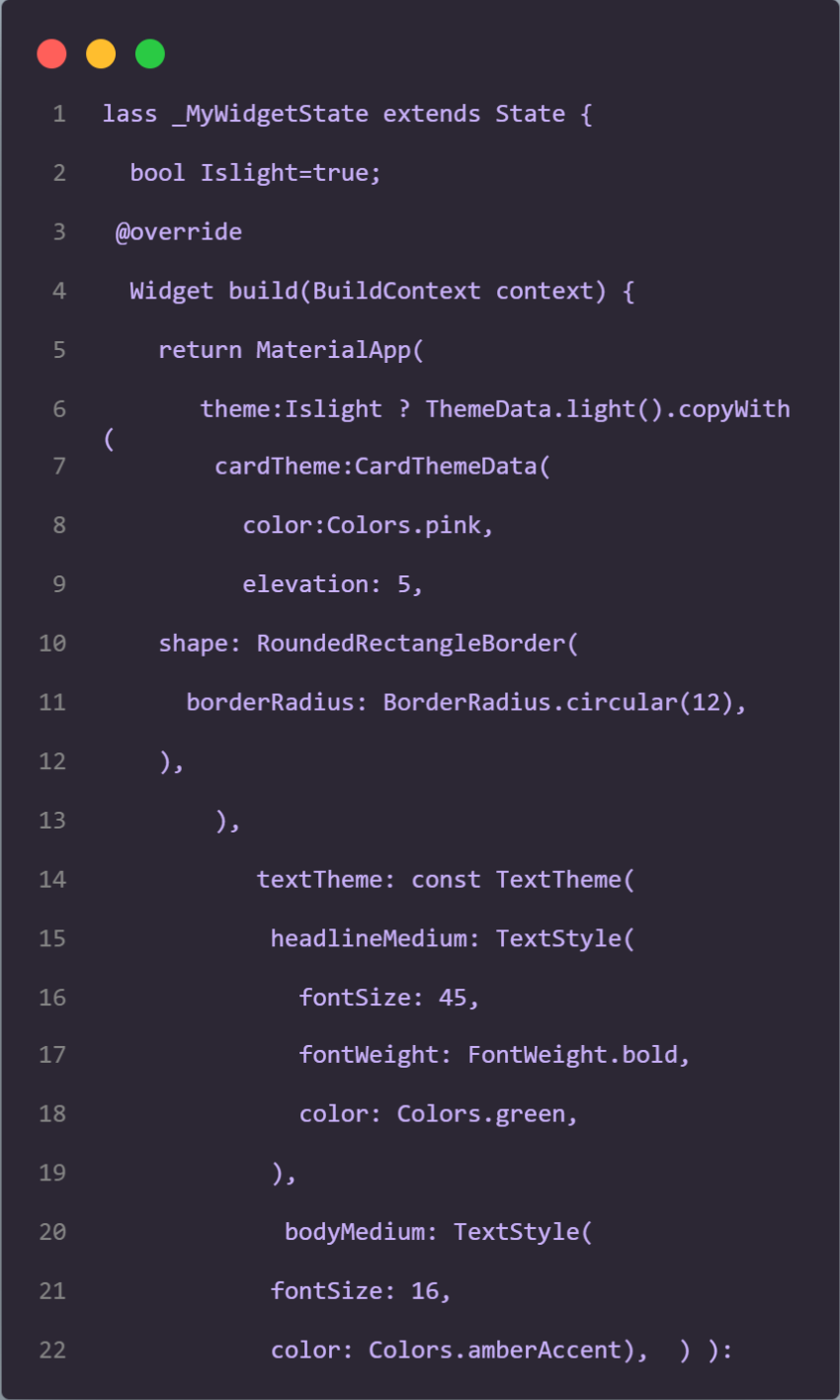
```

OUTPUT:



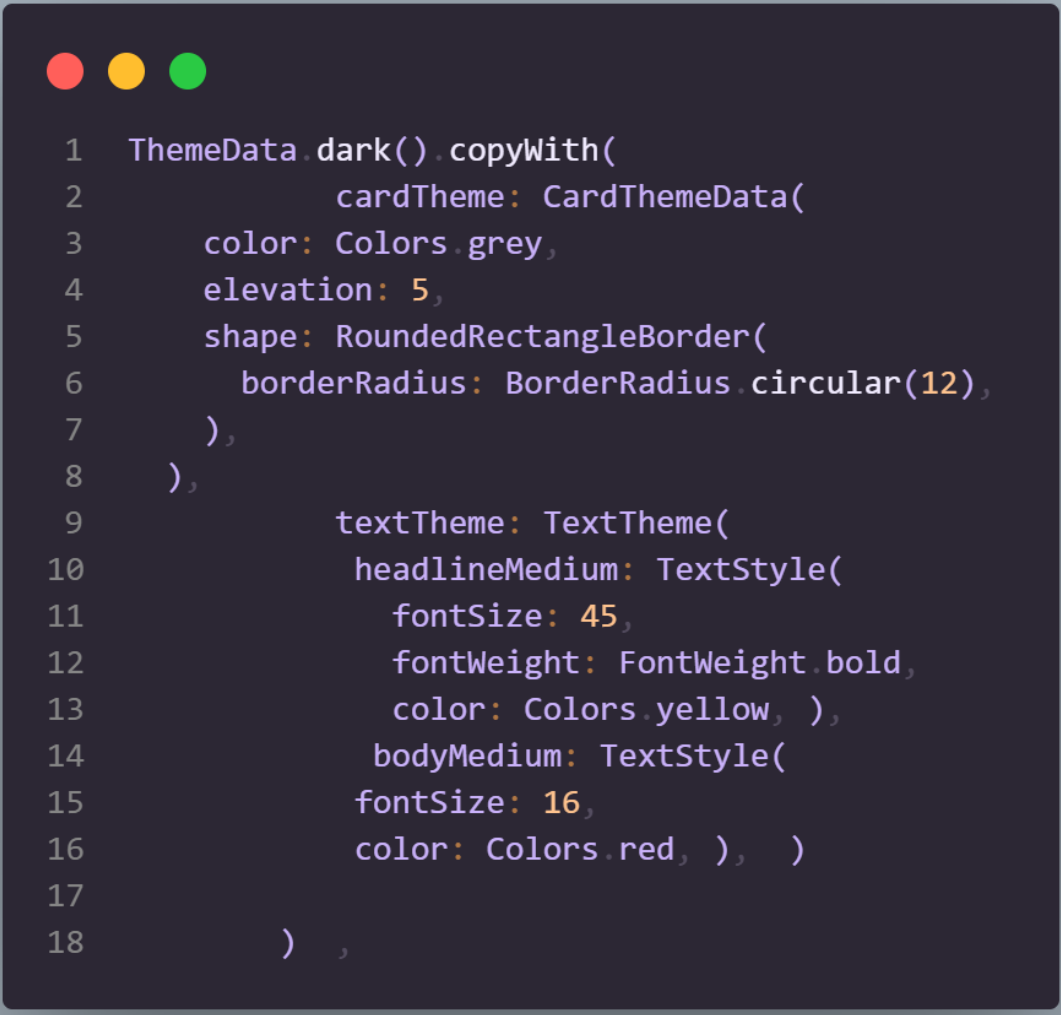
3. Apply Themes on cards, buttons and App bar.

Light theme cards:



```
1  class _MyWidgetState extends State {
2    bool Islight=true;
3    @override
4    Widget build(BuildContext context) {
5      return MaterialApp(
6        theme:Islight ? ThemeData.light().copyWith
7        (
8          cardTheme:CardThemeData(
9            color:Colors.pink,
10           elevation: 5,
11           shape: RoundedRectangleBorder(
12             borderRadius: BorderRadius.circular(12),
13           ),
14           textTheme: const TextTheme(
15             headlineMedium: TextStyle(
16               fontSize: 45,
17               fontWeight: FontWeight.bold,
18               color: Colors.green,
19             ),
20             bodyMedium: TextStyle(
21               fontSize: 16,
22               color: Colors.amberAccent), ) ) );
```

Dark theme cards:



```
1  ThemeData.dark().copyWith(  
2      cardTheme: CardThemeData(  
3          color: Colors.grey,  
4          elevation: 5,  
5          shape: RoundedRectangleBorder(  
6              borderRadius: BorderRadius.circular(12),  
7          ),  
8      ),  
9      textTheme: TextTheme(  
10         headlineMedium: TextStyle(  
11             fontSize: 45,  
12             fontWeight: FontWeight.bold,  
13             color: Colors.yellow, ),  
14         bodyMedium: TextStyle(  
15             fontSize: 16,  
16             color: Colors.red, ), ),  
17  
18     ) ,
```

OUTPUT:

